```
function createMap(countries, cities) {

  var width = 500;
  var height = 500;

  var projection = d3.geo.mollweide()
                     .scale(120)
                     .translate([width / 2, height / 2]);

  var geoPath = d3.geo.path().projection(projection);

  var featureSize = d3.extent(countries.features,
                    function(d) {return geoPath.area(d);});

  var countryColor = d3.scale.quantize()
                .domain(featureSize).range(colorbrewer.Reds[7]);

  d3.select("svg").selectAll("path").data(countries.features)
       .enter()
       .append("path")
       .attr("d", geoPath)
       .attr("class", "countries")
       .style("fill", function(d) {
                   return countryColor(geoPath.area(d))
       });

  d3.select("svg").selectAll("circle").data(cities)
      .enter()
      .append("circle")
      .attr("class", "cities")
      .attr("r", 3)
      .attr("cx", function(d) {return projection([d.x,d.y])[0]})
      .attr("cy", function(d) {return projection([d.x,d.y])[1]});
};
```

**For a Mollweide projection; shows the entire world**

**Measures the features and assigns the size classes to a color ramp**

**Colors each country based on its size**

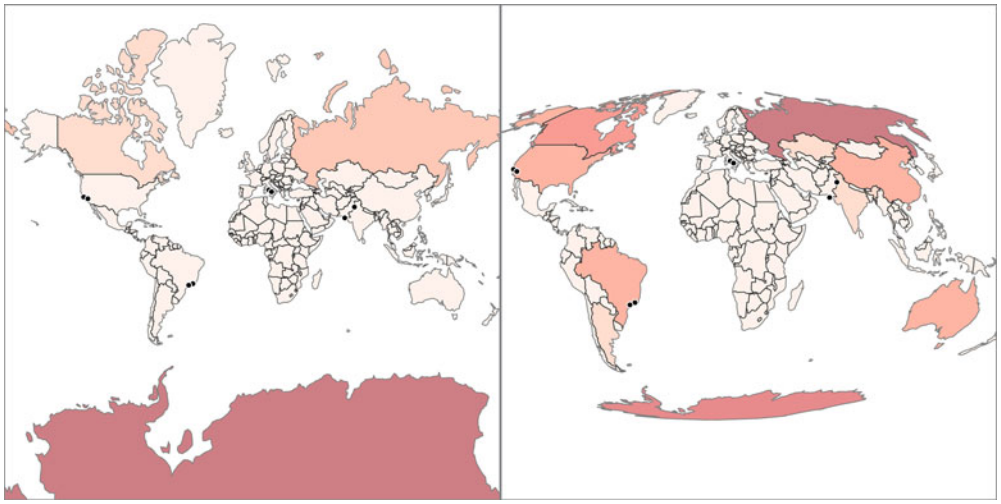


**Figure 7.6 Mercator (left) dramatically distorts the size of Antarctica so much that no other shape looks as large. In comparison, the Mollweide projection maintains the actual physical area of the countries and continents in your geodata, at the cost of distorting their shape and angle. Notice that `geo.path.area` measures the graphical area and not the actual physical area of the features.**